

scope, which allows the crew to view the docking target. The eight hydrogen peroxide thrusters located on the module are used to control the spacecraft's orientation, or attitude, during the descent until parachute deployment. It also has a guidance, navigation and control system to manoeuvre the vehicle during the descent phase of the mission.

This module weighs 2900 kilograms, with a habitable volume of 141 cubic feet. Approximately 50 kgs of payload can be returned to Earth in this module and up to 150 kgs if only two crewmembers are present. The Descent Module is the only portion of the Soyuz that survives the return to Earth.

iii) Instrumentation/Propulsion Module – This module contains three compartments: Intermediate, Instrumentation and Propulsion.

The intermediate compartment is where the module connects to the Descent Module. It also contains oxygen storage tanks and the attitude control thrusters, as well as the electronics, communications and control equipment. The primary guidance, navigation, control and computer systems of the Soyuz are in the instrumentation compartment, which is a sealed container filled with circulating nitrogen gas to cool the avionics equipment.

The propulsion compartment contains the primary thermal control system and the Soyuz radiator, which has a cooling area of 86 square feet. The propulsion system, batteries, solar arrays, radiator and structural connection to the Soyuz launch rocket are located in this compartment.

The propulsion compartment contains the system that is used to perform any manoeuvres while in orbit, including any rendezvous and the de-orbit burns necessary to return to Earth. The propellants are nitrogen tetroxide and unsymmetric-dimethylhydrazine. The main propulsion system and the smaller reaction control system, used for attitude changes while in space, share the same propellant tanks.

The two Soyuz solar arrays are attached to either side of the rear section of the Instrumentation/Propulsion Module and are linked to rechargeable batteries. Like the Orbital Module, the intermediate section of the Instrumentation/Propulsion Module separates from the Descent Module after the final deorbit maneuver and burns up in atmosphere upon re-entry.

Flights, Casualties and the End of Moonshot

On April 23, 1967, Soyuz 1 was launched into orbit. But a major controversy surrounds this particular launch, as it was suggested in the book *Starman* by Jamie Doran and Piers Bizony that Mishin and others knew it was not ready for flight and was a doomed mission. However, political pressure